

BUDYONNOVSK, Russian FederationWMO: **370610**Lat: **44.783N**Lon: **44.133E**Elev: **446**StdP: **14.46**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.7	7.0	-5.3	4.2	0.2	2.3	6.3	7.6	19.2	33.5	15.8	31.0	3.2	80	0.457

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.7	97.3	68.6	94.2	67.9	91.2	67.2	72.2	87.8	70.7	86.3	69.5	84.7	6.9	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.9	104.0	76.9	66.1	98.0	75.8	64.6	92.8	75.1	36.2	87.8	34.9	86.2	33.8	85.0	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	17.0	14.0	11.9	DB	-7.5	101.4	8.8	3.7	-13.9	104.0	-19.1	106.2	-24.0	108.2	-30.5	110.9
(5)				WB	-8.0	75.8	8.8	2.9	-14.3	77.9	-19.4	79.6	-24.3	81.2	-30.7	83.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.8	27.6	29.8	39.9	52.0	63.5	72.6	78.2	77.1	65.9	53.3	40.3	32.4
	DBStd	19.42	10.04	10.44	7.17	7.22	7.06	6.13	5.16	5.87	7.01	8.65	7.89	8.37
	HDD50	2551	696	564	319	59	3	0	0	0	1	65	299	545
	HDD65	5637	1161	984	777	394	114	12	0	1	72	370	742	1010
	CDD50	3587	0	0	7	118	421	677	873	840	477	167	7	0
	CDD65	1197	0	0	0	3	67	238	408	376	98	7	0	0
	CDH74	13104	0	0	1	54	696	2542	4719	4230	803	59	0	0
	CDH80	6006	0	0	0	8	182	1098	2405	2082	223	8	0	0
Wind	WSAvg	5.2	4.9	5.4	6.2	5.9	5.2	5.2	4.9	4.8	4.8	4.7	5.0	5.2
Precipitation	PrecAvg	17.3	1.1	0.9	1.3	1.6	2.0	2.2	1.5	1.7	1.4	1.4	1.2	1.2
	PrecMax	22.6	2.3	1.7	2.9	4.3	3.6	4.0	3.3	4.0	4.0	3.2	2.6	3.3
	PrecMin	11.0	0.1	0.4	0.4	0.1	0.2	0.4	0.2	0.1	0.2	0.1	0.2	0.6
	PrecStd	2.3	0.6	0.4	0.7	1.0	0.9	0.9	0.9	1.2	1.1	0.9	0.6	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.9	55.5	67.5	79.8	89.0	97.8	100.8	100.8	91.3	80.7	62.5	51.7
		MCWB	42.8	46.9	51.4	59.6	65.2	68.6	69.0	68.5	69.4	61.8	54.5	46.9
	2%	DB	43.0	48.1	60.3	74.4	84.7	93.2	97.7	97.1	85.4	73.9	56.8	47.5
		MCWB	39.9	42.8	48.8	57.6	63.3	67.1	68.4	68.9	65.7	60.3	51.9	44.4
	5%	DB	40.4	43.5	55.3	69.9	81.4	89.8	94.6	94.0	81.9	69.1	53.1	44.7
		MCWB	38.5	39.8	46.2	55.8	62.7	66.3	67.5	68.4	64.7	58.9	49.5	42.6
	10%	DB	38.2	40.9	51.2	65.6	77.9	86.3	91.5	90.5	78.6	65.1	50.5	42.3
		MCWB	36.6	38.4	44.3	53.8	61.5	65.4	66.9	67.9	63.3	57.5	47.9	40.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.6	47.2	53.6	61.5	68.3	72.1	74.0	74.7	72.5	64.3	55.5	47.6
		MCDB	47.0	54.0	64.0	75.7	82.9	90.3	91.0	89.4	85.2	73.5	60.6	50.1
	2%	WB	40.5	43.2	49.9	58.9	65.9	69.9	71.7	72.5	68.7	62.1	52.7	44.7
		MCDB	42.4	47.3	58.3	71.9	79.4	87.0	88.1	87.2	80.5	69.9	55.8	47.0
	5%	WB	38.7	40.4	47.3	56.8	64.4	68.4	70.2	71.0	66.6	60.3	50.2	42.8
		MCDB	40.2	42.8	53.9	67.7	77.4	83.9	86.1	86.1	78.2	67.3	52.5	44.5
	10%	WB	36.7	38.6	44.8	54.8	62.8	67.1	69.1	69.8	64.8	58.4	48.0	40.8
		MCDB	38.0	40.6	50.5	63.9	74.5	81.6	84.4	84.9	75.3	64.3	50.2	42.2
Mean Daily Temperature Range		MDBR	9.6	11.9	15.2	19.5	20.3	21.6	21.7	22.0	20.0	15.6	11.2	9.3
	5% DB	MCDBR	12.6	16.5	24.7	26.3	26.5	26.9	27.1	27.0	24.6	22.1	16.8	12.8
		MCWBR	9.4	11.5	13.9	11.9	9.7	8.2	7.2	8.1	8.6	10.4	11.4	9.7
	5% WB	MCDBR	11.1	15.3	22.1	23.1	22.8	23.6	23.3	23.2	21.7	19.1	14.2	11.7
		MCWBR	8.6	11.0	13.3	11.0	9.3	8.2	7.1	8.2	8.6	9.8	9.9	9.3
Clear-Sky Solar Irradiance	taub	0.346	0.380	0.418	0.470	0.437	0.442	0.466	0.472	0.441	0.395	0.362	0.352	
	taud	2.232	2.156	2.107	2.000	2.180	2.218	2.156	2.129	2.175	2.264	2.295	2.226	
	Ebn at Noon	234	247	254	251	265	264	255	248	245	240	228	217	
	Edh at Noon	28	36	44	53	45	44	46	46	41	32	26	26	
All-Sky Solar Radiation	RadAvg	352	615	931	1464	1834	2068	2033	1820	1329	778	411	285	
	RadStd	99	139	108	155	177	112	120	94	127	99	70	50	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (9 neighbors)	+0.65	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+243	N/A